# Talentra Database Design: Delivery Plan

**Northbridge Workforce Systems hackathon** | Engine: PostgreSQL 16/17 | Prepared Thu 15 Oct 2026

This document records the engine choice, the schema decisions, the build order, and a minute-by-minute time box for the first evaluation (Phase 1, Milestones 1-6). It is written to be handed to Claude Code as working context: section 9 lists what to generate first.

**Headline:** design before seed data. The first evaluation gates the signed Phase 1 design (entities, relationships, columns, keys and constraints for Master data, Profile, and Jobs and Applications). Seed data cannot be meaningful until columns and constraints are frozen, and the docket schedules it in Phase 5. The only seed work before the evaluation is a small smoke seed that proves the constraints reject bad data.

## 1. DB provider decision (B1): PostgreSQL

The docket's hardest rules map onto features PostgreSQL has natively and MySQL 8 does not.

| Docket requirement | PostgreSQL mechanism | MySQL 8 gap |
|---|---|---|
| Non-overlapping subscription periods (RG10), gig windows (B35), effective-dated recruiter mapping (B34) | EXCLUDE USING gist on a daterange / tstzrange with btree_gist | No exclusion constraints; triggers race under concurrent inserts |
| Privileges enforced in the DB: recruiter sees only own company's requirements; blocked party cannot see candidate | Row-Level Security policies keyed on current_setting('app.current_user_id') | No RLS; views only, easily bypassed |
| Soft deletes excluded by default; active-session counting at 10M scale | Partial indexes (WHERE deleted_at IS NULL, WHERE ended_at IS NULL) | No partial indexes |
| Job search and candidate de-duplication | tsvector + GIN; pg_trgm for fuzzy multilingual names | FULLTEXT is weaker; no trigram |
| Analytics marts and 100M applications | Native materialized views; declarative range partitioning | No materialized views |
| Append-only audit and ledgers; sensitive data | REVOKE UPDATE, DELETE + trigger guard; pgcrypto; column grants | Possible, but fewer primitives |
| Device limit under concurrent login (B45) | SELECT ... FOR UPDATE on seat row inside a transaction | Also possible (InnoDB) |

**Extensions used:** btree_gist, pg_trgm, pgcrypto, citext. **Collation:** UTF-8 database with ICU collation (und-x-icu) so multilingual names sort sensibly.

## 2. Conventions frozen before any design work

| Topic | Decision |
|---|---|
| Primary key | bigint GENERATED ALWAYS AS IDENTITY, named id, on every table. One type everywhere. |
| Natural keys | Unique constraints only: national ID token, legacy source ID, GSTIN/PAN, email where it identifies an account. |
| Foreign keys | Named <table>_id. Every FK states ON UPDATE / ON DELETE. Default: ON UPDATE NO ACTION, ON DELETE RESTRICT; CASCADE only for owned child rows (for example candidate_skill). |
| Schemas (modules) | mdm, profile, job, admin, org, gig, lic, src, billing, notif, dw |
| Names | snake_case, singular tables. Index idx_table_column; unique uq_table_column; check ck_table_rule; exclusion ex_table_rule; FK fk_table_column. |
| Physical objects | Partitions <table>_pYYYYMM; views v_; materialized views mv_; analytics dim_, fact_, agg_ in schema dw. |
| Enumerations | One lookup table per enum (application_status, employment_type ...) with code, label_key, is_active, sort_order. Real FKs, no free-text enums. |
| UI text | mdm.ui_label(key) + mdm.ui_translation(ui_label_id, language_id, text). Fallback: requested language, then en. |
| Person name | varchar(150), UTF-8. full_name plus optional given_name/family_name; no word-order assumption. |
| Email | citext, max 254, CHECK lowercase; unique where it identifies an account. |
| Phone | varchar(16) E.164 with CHECK (phone ~ '^\+[1-9][0-9]{6,14}$') plus country_id. |

| Topic | Decision |
|---|---|
| Money | numeric(14,2) plus currency_code char(3) FK to mdm.currency. Never float. |
| Percentage | numeric(5,2) with CHECK between 0 and 100. |
| Time | date for dates; timestamptz (UTC) for instants; source_timezone varchar(64) where display matters. |
| Audit columns | created_at, created_by, updated_at, updated_by on every business table; updated_at maintained by trigger. |
| Soft delete | deleted_at timestamptz on soft-deleted entities; default views filter it out. |
| Booleans | NOT NULL with a stated DEFAULT. |
| Unknown vs N/A | Disclosure code columns (for example salary_disclosure_code: disclosed, not_disclosed, not_applicable) with a CHECK tying it to NULL amounts. No 0, 'NA' or empty-string sentinels. |

```
-- Template every business table follows
CREATE TABLE job.job_requirement (
  id                     bigint GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  organisation_id        bigint NOT NULL REFERENCES org.organisation(id) ON DELETE RESTRICT,
  title                  varchar(200) NOT NULL,
  description            text NOT NULL,          -- unbounded: justified (job descriptions)
  description_language_id bigint REFERENCES mdm.language(id),  -- NULL = language not detected
  salary_disclosure_code varchar(20) NOT NULL REFERENCES mdm.salary_disclosure(code),
  salary_min             numeric(14,2),
  salary_max             numeric(14,2),
  currency_code          char(3) REFERENCES mdm.currency(code),
  is_open_to_consultant  boolean NOT NULL DEFAULT false,
  created_at timestamptz NOT NULL DEFAULT now(),  created_by bigint NOT NULL,
  updated_at timestamptz NOT NULL DEFAULT now(),  updated_by bigint NOT NULL,
  deleted_at timestamptz,
  CONSTRAINT ck_job_requirement_salary_range CHECK (salary_min IS NULL OR salary_max IS NULL OR salary_min <= salary
  CONSTRAINT ck_job_requirement_salary_disclosure CHECK (
    (salary_disclosure_code = 'disclosed' AND salary_min IS NOT NULL AND currency_code IS NOT NULL)
    OR (salary_disclosure_code <> 'disclosed' AND salary_min IS NULL AND salary_max IS NULL))
);
```

# 3. Schema decisions

## 3.1 Identity across modules

- One **admin.app_user** row per login identity; actor type comes from role assignment. Candidate, gig worker, recruiter and consultant profiles are 1:1 extensions keyed by app_user_id.
- Phase 1 needs FKs to users and organisations before their owning modules exist, so create **stub tables** (admin.app_user, org.organisation) with id, audit columns and status only. Phase 2 owners extend them; no FK needs rework.
- Credentials live in a separate admin.credential table (argon2id hash, restricted grants), never on app_user.

## 3.2 Applications and pipeline

- job.application holds current_status_code plus applied_at; unique (job_requirement_id, candidate_profile_id).
- job.application_status_history is append-only and records every transition with actor and reason.
- job.application_status_transition (from_code, to_code) lists legal moves; a trigger rejects anything else (for example applied straight to joined).
- Interview, offer and joining are child tables of application because they carry their own data (schedule, CTC, joining date). Rejection is a status plus a rejection_reason lookup.
- Partition application by range on applied_at (yearly at laptop scale, monthly at full scale); the unique key therefore includes applied_at or is enforced on a non-partitioned lookup table. Decide and record this in the justification doc.

## 3.3 Gig: separate subsystem (decision to defend)

Gigs get their own schema (gig.gig_request, gig_worker_profile, gig_availability, gig_application, gig_assignment, gig_handoff, gig_earning_reference, gig_worker_list_entry, gig_rating) and share identity, master data and analytics dimensions with the core.

| For separation | Cost of separation | Mitigation |
|---|---|---|
| Window, fixed/rate pricing, one-directional apply, external fulfilment, earnings and ratings differ on every axis. Sharing job_requirement would add many nullable columns and weaken CHECK constraints. | Unified pipeline reporting needs a union view; two application tables to maintain. | dw.fact_application carries a job_category dimension and is loaded from both; v_all_application unions the operational tables. Shared master data keeps skills and locations consistent. |

## 3.4 Constraint mechanisms (rules the panel will test)

| Rule | Mechanism |
|---|---|
| Subscription periods do not overlap per licence | EXCLUDE USING gist (licence_id WITH =, period WITH &&) on daterange |
| Gig windows do not overlap per worker | EXCLUDE USING gist (gig_worker_profile_id WITH =, work_window WITH &&) WHERE (status_code IN ('accepted','active')) |
| Recruiter-company mapping is effective-dated | EXCLUDE on (app_user_id, valid_period) so a recruiter maps to one company at a time |
| Device limit at login | In one transaction: SELECT seat row FOR UPDATE; count sessions WHERE ended_at IS NULL AND expires_at > now() (partial index); insert session or reject; write audit row either way. Release = set ended_at on logout, revoke, or expiry sweep. |
| Ledgers append-only | REVOKE UPDATE, DELETE from app roles plus a BEFORE UPDATE/DELETE trigger that raises. Corrections insert a new row with reverses_entry_id. |
| Wallet balance never negative | wallet.balance updated in the same transaction as the ledger insert, under row lock; CHECK (balance >= 0). |
| Recruiter sees own company's requirements only | RLS policy on job.job_requirement using the active recruiter mapping |
| Blocked party cannot see candidate | RLS on profile.candidate_profile: NOT EXISTS in admin.user_block for (candidate, viewer or viewer's organisation) |
| Consultant acts only on consultant-open requirements | Trigger or composite FK on src.sourcing_request checking job_requirement.is_open_to_consultant |
| Date ranges | CHECK (end_date IS NULL OR end_date >= start_date) on education, experience, projects, gigs |

## 3.5 Deletion, retention and sensitive data

- **Soft delete:** candidate_profile, job_requirement, organisation, gig_request, resume_version.
- **Hard delete:** junction rows (candidate_skill, job_requirement_skill, etc.).
- **Append-only:** audit_event, application_status_history, wallet_ledger_entry, payment_event, profile_view_event, profile_download_event; partitioned monthly.
- **Retention:** deletion on request anonymises Personal columns and sets deleted_at; a scheduled job purges after the stated window. Inactivity rule stated per entity in the data dictionary.
- **Sensitive class:** national and tax IDs stored as verification-service token plus masked last 4; credentials as argon2id hashes; contact details encrypted with pgcrypto, keys outside the database.
- **Out of scope noted:** employer sub-roles. Extension path: add organisation_id scope to role assignment and new permission rows; no other table changes.

# 4. Phase 1 entity lists (starting point for B4-B7)

| Module | Entities |
|---|---|
| Master data (mdm) | country, state, city, language, currency, timezone, skill, skill_category, qualification, qualification_level, field_of_study, institution, industry, job_role, job_function, employment_type, work_type (onsite/remote/hybrid), job_category (gig, staffing, professional, permanent, apprentice, campus, consultant, managed remote), salary_disclosure, notice_period, ui_label, ui_translation; status lookups: application_status, job_requirement_status, publish_state, rejection_reason, interview_mode, offer_status, proficiency_level |
| Stubs | admin.app_user, org.organisation (extended in Phase 2) |
| Profile | candidate_profile, candidate_contact, candidate_address, candidate_language, candidate_education, candidate_work_experience, candidate_skill, candidate_project, candidate_preference_location, candidate_preferred_role, resume_template, resume_version, assessment, assessment_attempt, course, course_enrolment, profile_completion_score, profile_view_event, candidate_activity_event, candidate_bookmark_event, referral |
| Jobs and Applications | job_requirement, job_requirement_location, job_requirement_skill, job_requirement_qualification, job_position (openings count and fill state), questionnaire, questionnaire_question, question_option, job_requirement_questionnaire, application, application_answer, application_status_history, application_status_transition, shortlist, shortlist_entry, interview, offer, joining, saved_search, job_search_event |

**Ownership notes:** shortlist and shortlist_entry belong to Jobs and Applications; the Sourcing module references them later. candidate_bookmark_event records history of bookmarks by employers/recruiters (Tier 2) and is append-only.

# 5. Time box for the first evaluation (3 hours)

Window: **11:30-14:30 IST**. Assumes four people: **Lead** (architect, owns conventions and DDL merge), **P2** (Master data), **P3** (Profile), **P4** (Jobs and Applications). With three people, Lead also takes Master data. With five, the fifth person builds the smoke seed and the compliance table from 12:45.

| Time (IST) | Milestone / backlog | Who | Output by end of slot |
|---|---|---|---|
| 11:30-11:45 | M1: B1-B3 provider, module list, assignments; freeze section 2 conventions | All | Decision note; conventions sheet; Postgres 16 running in Docker; git repo with /ddl /docs /seed |
| 11:45-12:10 | M2: B4, B6 identify entities | P2 mdm, P3 Profile, P4 Jobs; Lead writes stubs + extensions DDL | Three entity lists (start from section 4) |
| 12:10-12:20 | M2: B5, B7 validate entities | Cross-review: P3 reviews P4, P4 reviews P3, Lead reviews mdm | Validated entity lists; ownership conflicts resolved |
| 12:20-12:45 | M3: B8-B11 relationships | Same split | ERD in DBML (dbdiagram.io) with cardinality and ON DELETE rule per FK |
| 12:45-12:55 | Working lunch / buffer | All | Catch up on slippage |
| 12:55-13:35 | M4 + M5: B12-B19 columns and combined design | Same split; Lead enforces domains | Data dictionary rows: type, domain, nullability + reason, sensitive class |
| 13:35-14:05 | M6: B20-B23 keys and constraints | Each writes own DDL file; Lead merges and runs | ddl/00_ext.sql ... 03_job.sql apply cleanly on empty DB |
| 14:05-14:15 | Smoke tests | P4 + Lead | Section 6 tests pass; output captured |
| 14:15-14:25 | Compliance table + gig decision note + rehearsal | P2 compliance, P3 gig note, all rehearse | Section 7 table filled; 2-minute pitch |
| 14:25-14:30 | Submit | Lead | Signed Phase 1 design pack |

**Cut rules if behind:** at 13:00, if columns are not started, drop Tier 2 tables (assessment, course, referral, community) to entity-level only and note them as 'modelled at entity level, columns in Phase 4'. Never cut constraints, the compliance table, or the smoke tests; those are what a panel probes.

# 6. Smoke tests to show at the evaluation

Each should fail with a named constraint, proving rules live in the database.

```
-- 1. Salary range check            -> ck_job_requirement_salary_range
INSERT INTO job.job_requirement (..., salary_disclosure_code, salary_min, salary_max, ...) VALUES (..., 'disclosed',
-- 2. Sentinel instead of unknown   -> ck_job_requirement_salary_disclosure
INSERT ... salary_disclosure_code = 'not_disclosed', salary_min = 0 ...
-- 3. Date range                    -> ck_candidate_education_dates
INSERT INTO profile.candidate_education (..., start_date, end_date) VALUES (..., '2024-06-01', '2023-05-01');
-- 4. Duplicate application         -> uq_application_job_requirement_id_candidate_profile_id
-- 5. Illegal status jump           -> trigger: 'transition applied -> joined not allowed'
-- 6. Status history is append-only -> UPDATE profile/job history row raises
-- 7. Free-text enum                -> FK violation on application_status
-- 8. Uppercase email / bad phone   -> ck_app_user_email_lower / ck_candidate_contact_phone_e164
```

# 7. Rule compliance table (pre-filled for Phase 1)

| Rule group | Complies | Deviation and reason |
|---|---|---|
| 1 Naming | Yes | None. Module schemas are namespaces, not prefixes. |
| 2 Keys and constraints | Yes | Partitioned application: unique key must include partition key; recorded with the chosen workaround. |
| 3 Standard domains | Yes | Unbounded text only for job description, resume body; justified. |
| 4 Audit and lifecycle | Yes | created_by on stubs references app_user once Phase 2 lands; seed uses a system user id. |
| 5 Nullability | Yes | Every nullable column justified in the data dictionary. |
| 6 Sensitive classification | Yes | Class column on every data dictionary row. |
| 7 Multilingual | Yes | Language on user-entered content is nullable when undetected. |
| 8 Physical naming | Yes | Patterns in section 2. |
| Reserved keyword check | Yes | Avoided user, order, position, role, session, group: use app_user, payment_order, job_position, app_role, user_session, user_group. Check both engines' lists. |
| 9 Licensing, devices, sessions | Phase 3 | Mechanism stated in 3.4. |
| 10 Money, subscriptions, history | Phase 3 | Mechanism stated in 3.4. |

# 8. After the first evaluation: remaining milestones

| Phase | Milestones | Focus | Key risk |
|---|---|---|---|
| 2 | M7-M10 (B24-B35) | Administration, Organisation and account, Gig marketplace in parallel | Effective-dated recruiter mapping and gig window exclusion |
| 3 | M11-M14 (B36-B47) | Licences/devices, Sourcing, Subscriptions/wallet/download history | Device-limit locking under concurrency; ledger append-only |
| 4 | M15-M18 (B48-B59) | Notifications, analytics star schema, shared items (credentials, consent, audit, blocking) | High-volume delivery writes; SCD2 dimensions |
| 5 | M19 (B60-B64) | Seed: 1M profiles (duplicates, multilingual names), 100k jobs, 5M applications | Use COPY from generated CSV, drop/recreate indexes around load |
| 5 | M20-M21 (B65-B74) | 12-query set, EXPLAIN (ANALYZE, BUFFERS), p50/p95/p99, one index per module, extrapolation to 10M | Projection must rest on measured laptop numbers |
| 5 | M22-M23 (B75-B82) | PII notes, threat model, RLS/access tests, backup-restore timing, Gate 2 | RLS performance on candidate search |

**Seed generator (can start in parallel after 14:30):** Python with Faker locales (en_IN, hi_IN, ta_IN, ar, zh_CN) for multilingual names; inject 2-3% near-duplicate candidates (same email or phone variant, transliterated name); Zipf-skewed applications per job; status pipeline funnel such as applied 100%, shortlisted 30%, interviewed 12%, offered 4%, joined 3%. Load with COPY.

**Gaps in the docket:** Sections 8 (KPIs), 10 (performance method, RTO) and 12 are referenced but not in the issued PDF; request them before Phase 4/5. Phase 3 backlogs name Members A/B/C while the module table says 'Assigned

member'; confirm assignments with the organisers.

# 9. Using this document with Claude Code

Upload this PDF and the original docket, then run these prompts in order:

- "Using sections 2-4 of the plan, generate ddl/00_extensions.sql and ddl/01_mdm.sql for PostgreSQL 16, including lookup tables, ui_label and ui_translation, with every constraint named per section 2."
- "Generate ddl/02_stubs.sql, ddl/03_profile.sql and ddl/04_job.sql with all FKs, CHECK constraints, the status-transition trigger and append-only guards from section 3."
- "Generate docs/data_dictionary.csv with columns: schema, table, column, type, domain, nullable, null_reason, sensitive_class, delete_class."
- "Generate tests/smoke.sql implementing section 6, each test wrapped to report PASS when the expected error is raised."
- "Generate a DBML ERD for Phase 1 from the DDL."